

The Universal York Hamiltonian:

An Exact Scaling Law for Constant-Mean-Curvature Time in AdS₃ Gravity, and a Stress-Tensor Dilation Law on its Boundary

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Abstract

In three-dimensional gravity with a negative cosmological constant, the classical solution space is known completely: Mess showed that every globally hyperbolic AdS₃ spacetime on a closed surface of genus $g \geq 2$ corresponds to a point of $\mathcal{T}_g \times \mathcal{T}_g$, and every such spacetime carries a canonical constant-mean-curvature (CMC) time slicing, first proposed by York, whose Hamiltonian is simply the area of the slice. For each value of the CMC clock reading τ this construction hands us a different Hamiltonian H_τ on the same reduced phase space $T^*\mathcal{T}_g$, and there is no obvious reason these should be related to one another in any simple way. We show that they are: an exact, non-perturbative identity, $H_a(X, tq) = t^2 H_{at^2}(X, q)$, forces the entire one-parameter family of clock Hamiltonians to collapse onto a single universal profile function H_1 , sampled at a momentum scale that a CMC clock drives to infinity in a finite range of a natural angular variable. We derive this identity directly from the elliptic constraint equation that defines a CMC slice, establish the small-momentum expansion of H_1 in terms of the Weil–Petersson metric and curvature of Teichmüller space, and identify its large-momentum behaviour with the Thurston/Hubbard–Masur extremal-length geometry of quadratic differentials. We then take this exact bulk fact to the boundary. Because the momentum canonically conjugate to the boundary complex structure is, by the standard holographic dictionary, precisely the classical stress tensor of the dual CFT₂, the bulk scaling law becomes, semiclassically, a stress-tensor dilation law for the boundary generating functional: one universal boundary function, sampled at a stress-tensor scale set by the CMC clock reading. We state the sharp, falsifiable prediction this makes for the quartic curvature term of the boundary effective action, and we discuss how it sits alongside recent work identifying York time with the radial Hamiltonian flow dual to a $T\bar{T}$ -deformed CFT.

1 Introduction

Every theory of gravity we know how to write down as a Hamiltonian system runs into the same difficulty the moment it is quantised: general covariance turns the Hamiltonian into a constraint, and a wavefunction annihilated by a constraint does not evolve in any variable at all. This is the problem of time in canonical quantum gravity, first sharpened by DeWitt and Wheeler and given its modern statement by Kuchař and Isham: physical time must be extracted from the gravitational data itself, by singling out some combination of the fields and declaring it to be the clock, and different choices of clock are, a priori, entitled to disagree. It is one of the oldest open wounds in the subject, and for most of its history it has been debated in settings too complicated to resolve by direct computation.

Three-dimensional gravity with $\Lambda < 0$ is the one place where this needn’t be so. Because the classical solution space of 2+1-dimensional AdS gravity on a closed surface is completely and explicitly known, thanks to Mess, the question of what a clock does to the theory stops being a matter of principle and becomes a question one can simply answer, in closed form, using nothing more exotic than the differential geometry of Riemann surfaces. Among the many slicings one could choose for such a spacetime, one is distinguished by pure geometry rather than by any auxiliary structure: the constant-mean-curvature, or CMC, slicing proposed by York, in which the clock reading τ is literally the trace of the extrinsic curvature of the slice, its own instantaneous rate of expansion. The remarkable fact about this clock, known since the work of Moncrief and of Hosoya and Nakao in the late 1980s, is that the Hamiltonian which generates evolution in CMC time is, up to a constant, nothing but the area of the slice. Gravity’s dynamics, in this description, is the almost tautological statement that the universe’s own size changes as one

watches it change.

But this immediately raises a puzzle that is easy to state and, until now, has not been answered directly. For every value of τ the CMC construction hands us a Hamiltonian H_τ on the same reduced phase space, the cotangent bundle of Teichmüller space; and there is no a priori reason that H_τ and $H_{\tau'}$, for two different clock readings, should be related to each other by anything simpler than solving the Einstein equations twice. A naïve reading of the construction suggests an unbounded family of essentially unrelated classical Hamiltonian systems, one for every point on the clock’s dial, glued together only by the fact that they happen to generate flows on the same space of shapes.

This is where the present paper begins. Its first and central technical result is that this family is nowhere near as large as it looks. Buried inside the elliptic equation that defines a CMC slice is an exact scaling symmetry, holding without approximation and without appeal to any limit, that ties the Hamiltonian at momentum q and clock-dependent parameter a to the Hamiltonian at momentum tq and parameter at^2 , for every $t > 0$. Unwound, this identity says something almost embarrassingly simple: there is really only one Hamiltonian, a single function H_1 of the momentum alone, and every H_τ is nothing but this one function evaluated at a rescaled argument, the rescaling growing without bound as $\tau \rightarrow \pm\infty$ but doing so smoothly, in a finite range of a natural angular reparametrisation of time. The two asymptotic regimes that the literature on this subject usually treats as separate features of the theory, a Weil–Petersson quadratic response near the time-symmetric slice and a wholly different, Finsler-type, extremal-length behaviour far from it, turn out to be nothing more than the near-field and far-field behaviour of this one function. In a very concrete sense, the many clocks one could build out of CMC slicings were never many clocks at all.

That would already be a satisfying resolution of a fifty-year-old puzzle within its own corner of general relativity. But the fact that the momentum entering this scaling law is not an abstract symplectic variable, but is instead, via the standard machinery of $\text{AdS}_3/\text{CFT}_2$ holography, identifiable with the classical stress tensor of the boundary conformal field theory, means the story does not have to stop at the boundary of the bulk. The second aim of this paper is to carry the exact bulk scaling law across that boundary and show that it becomes, at the semiclassical level, a precise dilation law for how the boundary theory's own generating functional must depend jointly on the state's stress tensor and on the very same clock reading that parametrises the bulk construction. This gives a sharp, checkable prediction, stated at the end of the paper, for a coefficient in the boundary effective action that has not yet been computed independently; if the identification survives that test, an exact fact about how a wormhole breathes will have become an exact fact about how the dual CFT's stress tensor is organised.

The paper is organised as follows. Section 2 recalls Mess's parametrisation of the classical phase space and the CMC constraint equation. Section 3 develops the general notion of a clock as a choice of polarisation of this phase space, purely classically, and derives the existence of the clock Hamiltonian from elementary symplectic geometry. Section 4 solves the CMC constraint equation exactly and establishes the structural theorem for H_τ : well-posedness, monotonicity, the Weil–Petersson expansion at small momentum, and the extremal-length limit at large momentum. Section 5 is the technical heart of the paper: it derives the exact dilation law, the universal profile, the resulting Euler-type partial differential equation for H_τ , and the trigonometric substitution that compactifies the entire infinite range of CMC time into a finite angle. Section 6 carries all of this to the boundary, states the Stress-Tensor Dilation Theorem, and closes with the falsifiable prediction. Section 7 reflects on what has and has not been shown.

2 The classical phase space of AdS_3 gravity

2.1 Mess's parametrisation

Throughout, Σ_g is a closed, orientable surface of genus $g \geq 2$, and we set the AdS radius $\ell = 1$, so $\Lambda = -1$; Newton's constant G and the central charge $c = 3/2G$ appear only through the combination $16\pi G$. We write $\mathcal{T} = \mathcal{T}_g \cong \mathbb{R}^{6g-6}$ for the Teichmüller space of Σ_g , equipped with its Weil–Petersson (WP) Kähler metric g_{WP} and symplectic form ω_{WP} .

Theorem 2.1 (Mess; Moncrief; Hosoya–Nakao). *Let $\mathcal{Q} := \mathcal{T} \times \mathcal{T}$.*

- (a) *Maximal globally hyperbolic AdS_3 spacetimes on $\Sigma_g \times \mathbb{R}$ are in bijection with points $(\rho_L, \rho_R) \in \mathcal{Q}$, the left- and right-moving hyperbolic structures induced at the two ends of the spacetime, with $\omega_{\mathcal{Q}} = \lambda_G(\omega_{WP}^L - \omega_{WP}^R)$ for a fixed constant λ_G .*
- (b) *Each such spacetime carries a unique foliation by constant-mean-curvature surfaces Σ_τ , labelled by $\tau =$*

$-\text{tr } K$.

- (c) *Writing $X_\tau \in \mathcal{T}$ for the conformal class induced on Σ_τ and q_τ for the holomorphic quadratic differential built from its traceless second fundamental form, the map $\Phi_\tau : (\rho_L, \rho_R) \mapsto (X_\tau, q_\tau)$ is a symplectomorphism $\mathcal{Q} \rightarrow T^*\mathcal{T}$.*
- (d) *Time reversal $\sigma(\rho_L, \rho_R) = (\rho_R, \rho_L)$ satisfies $\Phi_0 \circ \sigma = \iota \circ \Phi_0$ with $\iota(X, q) = (X, -q)$, so the time-symmetric, Fuchsian locus $\rho_L = \rho_R$ is sent to the zero section of $T^*\mathcal{T}$.*

It is worth pausing on what this theorem actually says, because the two holonomies ρ_L, ρ_R are easy to misread as data on a single slice. They are not: a point of $\mathcal{Q}$ is an entire spacetime, specified by the two asymptotic hyperbolic shapes it interpolates between at the far past and far future of conformal infinity, and Mess's theorem says that this is *all* the data there is. Given any pair of hyperbolic structures on Σ_g , there is exactly one maximal globally hyperbolic AdS_3 spacetime realising them as its two conformal boundaries, and every classical solution arises this way. This is what makes the theory tractable: the classical phase space is not merely known to exist, it is a completely explicit, finite-dimensional, well-studied object from Teichmüller theory.

Part (c) trades this description for a more dynamical one, and it is the one we will use throughout. The CMC parameter $\tau = -\text{tr } K$ measures the local expansion rate of a slice: $\tau > 0$ means the slice is instantaneously growing, $\tau < 0$ means it is contracting, and $\tau = 0$ singles out the unique momentarily static slice, the “waist” of the wormhole, which by part (d) is exactly the time-symmetric, Fuchsian locus. As τ sweeps from $-\infty$ to $+\infty$, the CMC foliation carries us from the wormhole's contracting past, through its waist, to its expanding future, and (X_τ, q_τ) records, at each instant, the conformal shape of the universe and the shear, gravitational-wave-like content riding on top of it.

2.2 The shape of a CMC slice

Fixing a conformal representative ρ (the hyperbolic metric in the class X) and a holomorphic quadratic differential $q \in H^0(K_X^2)$, define

$$\beta := \rho^{-2}|q|^2, \quad \mu := \bar{q}/\rho, \quad a := 1 + \tau^2/4.$$

The quantity β is a genuine function on Σ_g (it does not depend on the choice of local coordinate), and it measures the local density of shear at a point of the surface; $a \geq 1$ is simply a repackaging of the clock reading that will turn out, in Section 5, to be exactly the right variable in which the theory becomes transparent.

Lemma 2.2 (CMC constraint equation). *A metric $h = e^{2\psi}\rho$ on Σ_g arises as the induced metric on a CMC- τ slice with shear data q if and only if ψ solves*

$$\Delta\psi = a e^{2\psi} - 1 - \frac{1}{2}\beta e^{-2\psi}, \quad (\text{E})$$

where Δ is the Laplacian of ρ .

Proof. The Hamiltonian constraint of 2+1 gravity, $R^{(2)} + (\text{tr } K)^2 - K_{ij}K^{ij} = -2$, together with the momentum constraint, which forces the traceless part B_{ij} of the second

fundamental form to be (the real part of) a holomorphic quadratic differential q , reduces directly to (E) once the metric is written conformally as $h = e^{2\psi}\rho$. $\square$

Equation (E) is a Liouville-type equation: the first two terms on the right are exactly what one would write down to force a constant-curvature metric in the conformal class of ρ , and the last term is a source proportional to $|q|^2$, the local density of shear. It is worth dwelling on the physical content of this fact, because it is easy to let the formalism obscure it: finding a CMC- τ slice with prescribed shear q is, mathematically, exactly the same problem as finding a constant-curvature metric on a surface in the presence of a matter source built from the wave content q itself. The gravitational waves riding on a CMC slice curve the slice they ride on, in precisely the sense that ordinary matter curves an ordinary Riemannian manifold.

Remark 2.3 (Normalisation). With β and μ defined directly from q as above, one has the exact identity $\beta dA_\rho = |\mu|_\rho^2 dA_\rho$, so that $\int_{\Sigma_g} \beta dA_\rho = \|\mu\|_2^2$ with coefficient exactly one. No further normalisation constant needs to be introduced by hand, and every coefficient appearing later, in particular in Theorem 4.1(iii) and in the holographic matching of Section 6, is a genuine number rather than one defined only up to an unfixed rescaling.

2.3 The reduced action and the York Hamiltonian

Proposition 2.4 (Reduced action). *Let $p := B/16\pi G$, where B is the traceless second fundamental form built from q . Then the Einstein–Hilbert action, once the constraints have been solved, reduces to*

$$S_{\text{red}} = \int (\langle p, \dot{X} \rangle - H_\tau) d\tau, \quad H_\tau(X, q) = \frac{A_\tau(X, q)}{16\pi G}, \quad (1)$$

where A_τ is the area of the CMC- τ slice in the metric h solving (E).

This is the fact around which the whole paper turns, so it is worth stating in words as plainly as possible. Ordinary Hamiltonian mechanics attaches a single number, the energy, to a configuration; here, the number attached to an entire spatial slice of the universe is a purely geometric quantity, the slice’s own area, and the dynamics of gravity is exactly the statement that this area changes as one slice is traded for a neighbouring one. There is no separate “matter content” driving the evolution: the area itself, as a function of the shape X and the shear q carried on that shape, is the whole of the dynamics.

3 Clocks as choices of polarisation

Before solving (E) explicitly, it is worth setting up the general apparatus of a “clock” at the level of symplectic geometry, both because it clarifies what is special about the CMC choice and because it is this apparatus that produces H_τ as a genuine Hamiltonian rather than merely as a suggestively named quantity.

3.1 Clock atlases

Definition 3.1 (Clock atlas). Let (Q, ω) be a symplectic manifold and Q a manifold. A *clock atlas* is a smooth

family of symplectomorphisms $\Phi_\tau : Q \rightarrow T^*Q$. Its polarisation at reading τ is the distribution $\mathcal{F}_\tau = \ker d(\pi_Q \circ \Phi_\tau)$: the directions in Q that the clock, at that instant, treats as not changing the reading.

The York atlas of Theorem 2.1(c), with $Q = \mathcal{T}$ and $\Phi_\tau = (X_\tau, q_\tau)$, is the running example. Physically, a polarisation is a choice of which half of the gravitational phase-space data one is willing to call “the shape of the universe right now” and which half one calls “its instantaneous rate of change.” Different clocks make this split differently; the entire content of the formalism below is that the split, and nothing else, is what generates a Hamiltonian.

Proposition 3.2 (Existence of the clock Hamiltonian). *If $H^1(T^*Q; \mathbb{R}) = 0$, as holds for $Q = \mathcal{T}$ since Teichmüller space is contractible, then for every clock atlas there is a smooth function $H_\tau \in C^\infty(T^*Q)$, unique up to the addition of a function of τ alone, such that $\partial_\tau \Phi_\tau = X_{H_\tau} \circ \Phi_\tau$.*

Proof. Set $V_\tau := (\partial_\tau \Phi_\tau) \circ \Phi_\tau^{-1}$, a time-dependent vector field on T^*Q . Since each Φ_τ is symplectic, $\mathcal{L}_{V_\tau} \omega_{\text{can}} = 0$, so $\iota_{V_\tau} \omega_{\text{can}}$ is closed; contractibility of T^*Q makes it exact, and H_τ is (minus) its primitive. $\square$

This is nothing more than the familiar fact, from elementary classical mechanics, that any time-dependent canonical transformation is generated by some time-dependent Hamiltonian. What is specific to gravity is not this proposition, which is pure symplectic geometry and holds for any clock atlas whatsoever, but the identification, in Proposition 2.4, of the resulting H_τ with an honest, computable geometric quantity: the area of the slice.

3.2 Time as non-commutativity

A clock that changes with τ carries a further piece of purely classical structure, worth recording because it gives an operational meaning to the idea that different clock readings disagree about “now.”

Proposition 3.3 (The clock tensor). *In Darboux coordinates (x^i, p_i) on T^*Q ,*

$$\{x_\tau^i, x_{\tau'}^j\}_Q = (\tau' - \tau) \left(\frac{\partial^2 H_\tau}{\partial p_i \partial p_j} \right)_\tau + O((\tau' - \tau)^2),$$

so that x_τ^i and $x_{\tau'}^j$ fail to Poisson-commute at first order in $\tau' - \tau$, with coefficient the Hessian of H_τ in the momenta.

Proof. Expand $x_{\tau'}^j = x_\tau^j + (\tau' - \tau) \partial H_\tau / \partial p_j + O((\tau' - \tau)^2)$ using Proposition 3.2, take the Poisson bracket with x_τ^i , and use $\{x^i, \partial H_\tau / \partial p_j\} = \partial^2 H_\tau / \partial p_i \partial p_j$. $\square$

Read operationally: x_τ^i and $x_{\tau'}^j$ are the shapes that two clock readings, τ and τ' , would each call “the shape of the universe now,” read off the same underlying space-time. That they fail to Poisson-commute is the classical shadow of the fact that these two notions of “now” are not independent pieces of data but are tied together by the dynamics: a spacetime does not have a shape and, separately, a rate of change of that shape; it has a single history, and different clocks slice that history differently. The Hessian $\partial^2 H_\tau / \partial p_i \partial p_j$ measures precisely how

fast this disagreement grows as the two readings separate, and it is a purely classical, gauge-invariant quantity, with no reference to $\hbar$ anywhere in its definition. We will not need to quantise anything to extract physics from it: it already tells us, at the classical level, how much two nearby clocks disagree about the present.

4 The exact York Hamiltonian

We now solve the CMC constraint equation (E) in closed form, in the precise sense of establishing everything one needs to know about the solution without ever writing it down explicitly.

4.1 The variational problem

On $H^1(X)$, define

$$S_a[\psi] = \int_X \left[\frac{1}{2} |\nabla \psi|^2 + \frac{a}{2} e^{2\psi} - \psi + \frac{1}{4} \beta e^{-2\psi} \right] dA_\rho$$

$$S^*(a) := \min_{\psi} S_a[\psi].$$

Equation (E) is exactly the Euler–Lagrange equation of S_a . The first two terms of the integrand are the ordinary Liouville functional, whose minimiser puts a constant-curvature metric on the conformal class of ρ ; the last term is the potential-energy cost of carrying the shear q . Let $G := (2 - \Delta)^{-1}$ and $\mathcal{R}(\mu) := 4(|\mu|^2, G|\mu|^2)_{L^2} > 0$.

Theorem 4.1 (The exact York Hamiltonian). (i)

Well-posedness. For every (X, q, τ) , equation (E) has a unique smooth solution, the minimiser of S_a .

(ii) **Structure.** $H_\tau = A/16\pi G$ is real-analytic in (q, τ) , satisfies $H_\tau = \frac{1}{8\pi G} \partial_a S^*$, and

$$\partial_a A = -2 \left\langle e^{2\psi}, (-\Delta + 2ae^{2\psi} + \beta e^{-2\psi})^{-1} e^{2\psi} \right\rangle < 0;$$

moreover $H_\tau(X, q) = H_\tau(X, -q) = H_{-\tau}(X, q)$.

(iii) **Small-momentum expansion.** As $q \rightarrow 0$,

$$16\pi G H_\tau = \frac{\text{Vol}}{a} + \|\mu\|_2^2 - \frac{a}{2} \mathcal{R}(\mu) + O(\|q\|^6), \quad (2)$$

where $\mathcal{R}(\mu) = -\text{Rm}_{WP}(\mu, \bar{\mu}, \mu, \bar{\mu}) = -\text{HolK}(\mu) \|\mu\|_2^4$, the holomorphic sectional curvature of the Weil–Petersson metric in the direction μ .

(iv) **Extremal-length limit.** As $|\tau| \rightarrow \infty$ with μ held fixed, $|\tau| H_\tau \rightarrow \|\mu\|_1 / 8\pi G$, proportional to the extremal length of the singular foliation associated to q by Hubbard and Masur.

Proof sketch. (i) The function $F(\psi) = ae^{2\psi} - 1 - \frac{1}{2}\beta e^{-2\psi}$ is strictly increasing, so a standard super/sub-solution and monotone-iteration argument gives a smooth solution, the maximum principle gives uniqueness, and S_a is manifestly strictly convex. (ii) Invertibility of $\Delta - F'(\psi)$ gives real-analyticity by the analytic implicit function theorem; the envelope theorem applied to $S^*(a)$ gives the first identity, and differentiating (E) in a gives the monotonicity. (iii) Expand $\psi = \psi_0 + \varepsilon_1 + \dots$ in powers of q ; the linearised equation $(\Delta - 2)\varepsilon_1 = -\frac{1}{2}\beta$ and the Tromba–Wolpert identification of the quartic term with the Weil–Petersson curvature tensor give (2). (iv) Sandwiching $S^*(a)$ between

explicit trial functions and using the concavity of S^* in a gives the stated limit, which is then recognised as (a multiple of) the L^1 -norm defining extremal length. $\square$

Each part of this theorem has a transparent physical reading, and it is worth walking through all four before moving on, since the paper’s later results are read off this structure almost without further computation. Part (i) says that a CMC slice with prescribed expansion rate and prescribed shear content always exists and is unique: there is exactly one way to embed a “shape carrying a given amount of gravitational-wave content” into an AdS_3 wormhole at a given instant. Part (ii) says the area of the slice shrinks monotonically as $|\tau|$ grows in either time direction, which is the CMC-time avatar of the familiar big-bang/big-crunch picture of a wormhole pinching down to the past and to the future; and the identity $H_\tau = \frac{1}{8\pi G} \partial_a S^*$ says the area is literally a thermodynamic derivative of a free energy $S^*(a)$, with a playing the role of an inverse temperature. Part (iii) says that near the time-symmetric slice, the energy cost of carrying a small amount of shear is a positive-definite quadratic form, the Weil–Petersson metric itself, corrected by a quartic self-interaction set by the curvature of Teichmüller space; this is exactly the shape one would write down for the leading terms of an effective Hamiltonian for a weakly self-interacting field. Part (iv) says that this quadratic, Riemannian picture is only the near-symmetric approximation to something else entirely: far from time-symmetry, the natural notion of the “size” of the shear degenerates from a Euclidean-looking norm into a genuinely different, Finsler-type, degree-one norm tied to the topology of the singular foliation that q defines on the surface. This is a change of geometry every bit as qualitative as the transition from a weak-field to a strong-field regime in ordinary general relativity, except that here it takes place entirely within the reduced classical mechanics of a fixed and otherwise unremarkable model.

4.2 An illustrative check

The coefficients appearing in (2) depend only on the operator $-\Delta + 2$ evaluated at the background solution ψ_0 and on the elementary fact that $Gf = \frac{1}{2} \int f$ for constant f . As a purely algebraic sanity check, one can minimise S_a numerically on a flat 2π -periodic torus, using a truncated Fourier expansion with 64^2 modes and a shear profile $\beta = \epsilon^2 f^2$ for a fixed test function f . Doing so reproduces the predicted quartic coefficient in (2) to within half a percent at $\epsilon = 0.05$, with the discrepancy scaling as ϵ^6 as it should; the ratio $A\sqrt{2a}/\int \sqrt{\beta}$, which should tend to 1 in the limit of part (iv), does so as a is taken up to 10^5 . This is only an algebraic consistency check on a flat background, not a computation on an actual hyperbolic surface, but it confirms that no sign or factor has gone astray in the derivation above.

5 The universal profile: an exact Euler equation for time

We now come to the paper’s central technical discovery: buried inside Theorem 4.1 is an exact, non-perturbative scaling symmetry that collapses the entire one-parameter

family $\{H_\tau\}$ onto a single function.

Proposition 5.1 (Dilation law). *For all $t > 0$,*

$$H_a(X, tq) = t^2 H_{at^2}(X, q).$$

Proof. Substituting $\psi = \psi' + \ln t$ into S_a and rescaling $\beta \rightarrow t^2 \beta$ gives $S_a^*(t^2 \beta) = S_{at^2}^*(\beta) - \text{Vol} \cdot \ln t$; differentiating in a and using $H_a = \frac{1}{8\pi G} \partial_a S_a^*$ gives the stated identity. $\square$

The content of this identity is that rescaling the shear content of a slice by a factor t can always be exactly compensated by rescaling the clock's own parameter $a \mapsto at^2$: a scaling symmetry that ties together the size of the momentum and the reading of the clock, in the spirit of a renormalisation-group flow with $\sqrt{a}$ playing the role of the flow's scale.

Corollary 5.2 (The universal profile). *There is a single function $H_1(X, \cdot)$ on $T_X^* \mathcal{T}$ such that*

$$H_a(X, q) = \frac{1}{a} H_1(X, \sqrt{a} q) \quad (a \geq 1). \quad (3)$$

Proof. Set $t = \sqrt{a'}$ and evaluate Proposition 5.1 at $a = 1$. $\square$

This is the fact promised in the introduction, stated with full precision: there is no genuine family of Hamiltonians at all. There is a single fixed function H_1 of the momentum, and the entire clock-dependence of the theory is the purely kinematic act of asking where on the graph of H_1 a given clock reading happens to be looking. Every CMC clock, at every reading, is reading off the same underlying function; only the scale at which it samples that function changes.

Proposition 5.3 (Euler equation for time). *H_τ satisfies the inhomogeneous Euler partial differential equation*

$$a \partial_a H_a - \frac{1}{2} q \cdot \nabla_q H_a + H_a = 0. \quad (4)$$

Proof. Differentiate Proposition 5.1 in t at $t = 1$: the left side gives $q \cdot \nabla_q H_a$, the right side gives $2H_a + 2a\partial_a H_a$, and rearranging gives (4). $\square$

Equation (4) is not a statement of homogeneity, which would simply read $a\partial_a H = 0$ and would in fact be false; it says that the failure of H_τ to be independent of a is controlled entirely by its own radial derivative with respect to the momentum. This is the algebraic fingerprint of a function that secretly depends on its two arguments only through one combined ray variable, which is exactly what Corollary 5.2 exhibits directly. The same fingerprint occurs whenever a physical quantity depends on two parameters only through their dimensionless ratio, as happens near a critical point in statistical mechanics; here the combination is forced on us by nothing more exotic than the exact classical dynamics of the CMC constraint.

Proposition 5.4 (Trigonometric reduction of CMC time). *Let $\theta = \arctan(\tau/2) \in (-\frac{\pi}{2}, \frac{\pi}{2})$, so that $\tau = 2 \tan \theta$ and $a = \sec^2 \theta$. Then*

$$H_\tau(X, q) = \cos^2 \theta H_1(X, \sec \theta \cdot q). \quad (5)$$

Proof. Substitute $a = \sec^2 \theta$ into (3) and use $1/a = \cos^2 \theta$, $\sqrt{a} = \sec \theta$. $\square$

Writing $\Xi(\theta) := \sec \theta \cdot q$, a CMC history through (X, q) reads, at clock-angle θ , the fixed profile H_1 evaluated at $\Xi(\theta)$ and rescaled by $\cos^2 \theta$; for every $q \neq 0$, $\|\Xi(\theta)\| \rightarrow \infty$ as $\theta \rightarrow \pm \frac{\pi}{2}$, that is, as $\tau \rightarrow \pm \infty$. The clock reaches “the end of time” by driving the argument of a fixed function to infinity in a finite range of angle, not by that function itself changing character. This substitution is not a convenience chosen for its own sake: $\tau = 2 \tan \theta$ is exactly the familiar conformal compactification of AdS global time, the same device that turns the infinite proper-time range of a bulk observer into the finite range of a Penrose diagram, with the conformal boundary reached precisely at $\theta = \pm \pi/2$. Here θ does double duty, serving simultaneously as the natural AdS conformal-time coordinate and as the coordinate in which a diverging clock reading becomes a finite parameter, which is exactly what one would want from any coordinate designed to make honest sense of an unbounded clock.

Corollary 5.5 (One function, two regimes). *Substituting $q = \Xi/\sqrt{a}$ into (2) and using the quartic homogeneity of $\mathcal{R}$,*

$$16\pi G H_1(X, \Xi) = \text{Vol} + \|\mu_\Xi\|_2^2 - \frac{1}{2} \mathcal{R}(\mu_\Xi) + O(\|\Xi\|^6) \quad (\Xi \rightarrow 0), \quad (6)$$

independent of a , as it must be; and substituting the extremal-length asymptotics of Theorem 4.1(iv) into (3) with $a \approx \tau^2/4$ gives

$$16\pi G H_1(X, \Xi) \rightarrow \|\Xi\|_1 \quad (\|\Xi\| \rightarrow \infty). \quad (7)$$

Proof. For (6), substitute $q \rightarrow \Xi/\sqrt{a}$, $\mu \rightarrow \mu_\Xi/\sqrt{a}$ into (2), use $\mathcal{R}(\mu_\Xi/\sqrt{a}) = \mathcal{R}(\mu_\Xi)/a^2$, multiply through by a , and use $H_1(X, \Xi) = aH_a(X, \Xi/\sqrt{a})$: every explicit power of a cancels. For (7), substitute $|\tau| \approx 2\sqrt{a}$ and $\|\mu_\Xi\|_1 = \sqrt{a}\|\mu\|_1$ into the limit of Theorem 4.1(iv). $\square$

The Weil–Petersson regime and the Teichmüller–Finsler regime of the theory, usually treated in the literature as two separate facts about 2+1-dimensional AdS gravity, are therefore not two phenomena at all: they are simply the near-origin and far-field behaviour of one fixed function $H_1(X, \cdot)$ on the vector space $T_X^* \mathcal{T}$, and the crossover between them is nothing but the scale at which these two asymptotics of the same function cease to agree. The analogy worth keeping in mind is the relation between weak-field and strong-field gravity itself: ordinary general relativity reduces to Newtonian gravity for weak fields while exhibiting qualitatively different behaviour, horizons among them, in the strong-field regime, and yet remains, throughout, one and the same theory. In exactly this sense, H_1 looks Riemannian for small momenta and genuinely Finsler for large ones while remaining, throughout, one and the same function.

6 The boundary reading: a stress-tensor dilation law

We now take the exact bulk result of Section 5 to the conformal boundary. Nothing in what follows adds a new bulk computation; the entire content of this section is that the

scaling law already proved above, once read through the standard holographic dictionary, is not merely a fact about how a CMC slice breathes, but a precise statement about how the dual CFT_2 's own generating functional must organise itself.

6.1 The holographic dictionary for CMC data

The starting point is a fact that is standard in $\text{AdS}_3/\text{CFT}_2$ and needs no new argument here: the momentum $p = B/16\pi G$ conjugate to the boundary complex structure X in the reduced action (1) is exactly the (holomorphic component of the) Brown–York boundary stress tensor of the spacetime. This was established for AdS gravity in general by Brown and York, made concrete and finite for asymptotically AdS spacetimes by Balasubramanian and Kraus, and worked out in precisely the Teichmüller-theoretic language used in this paper by Scarinci and Krasnov, who show explicitly that the holographic charges of an AdS_3 spacetime, its mass and angular momentum among them, are periods of the very quadratic differential q that appears in Theorem 2.1(c). Since $B_{zz} = q$ by Lemma 2.2, this means that q is not merely some abstract cotangent coordinate on the CMC phase space; up to the fixed, computable normalisation supplied by holographic renormalisation, it already *is* the classical stress tensor T of the dual CFT_2 , evaluated in the state and background complex structure that the CMC slice describes.

This identification has a natural home in the representation theory of the boundary symmetry algebra. A traceless, holomorphically conserved stress tensor on a Riemann surface is, by definition, a holomorphic quadratic differential, so $T(z) dz^2 \in H^0(K_X^2)$ is not an analogy but a tautology; and the space of such objects modulo the inhomogeneous Schwarzian shift under which T transforms is precisely a coadjoint orbit of the Virasoro group, in the sense of Witten and of Alekseev and Shatashvili. The bulk CMC construction, in this light, is producing an explicit family of points on such an orbit, parametrised by the clock reading τ , together with an exact formula for how the gravitational free energy attached to that orbit point depends on where on the orbit one sits.

6.2 The Stress-Tensor Dilation Theorem

Granting this dictionary, Corollary 5.2 translates immediately into a statement purely about the boundary.

Theorem 6.1 (Stress-tensor dilation). *Let $\mathcal{E}(X, T) := 16\pi G H_1(X, T)$, with T identified with q under the holographic dictionary above. Then, exactly and without approximation on the bulk side,*

$$16\pi G H_\tau(X, T) = \frac{1}{1 + \tau^2/4} \mathcal{E}\left(X, \sqrt{1 + \tau^2/4} T\right). \quad (8)$$

At the semiclassical (large central charge, $c = 3/2G \rightarrow \infty$) level, at which the on-shell bulk action computes the leading saddle of the boundary CFT's generating functional in a heavy state with stress tensor T , this says that the entire family of CMC-time free energies collapses onto a single boundary functional $\mathcal{E}(X, \cdot)$, sampled at a stress-tensor scale $\sqrt{1 + \tau^2/4} T$ that grows with the clock reading.

It is worth being precise about exactly what is proved and what is being read into it, since it is easy to let

the elegance of (8) obscure the join between the two. The equation itself is Corollary 5.2, relabelled; nothing about it is new mathematics. What is new is the physical claim that $\mathcal{E}$, evaluated on this exact bulk quantity, computes a genuine semiclassical CFT quantity: the Legendre transform of the CFT's generating functional with respect to the boundary complex structure, evaluated at fixed stress tensor T rather than at fixed background metric. This claim rests entirely on standard, independently established holography, not on anything argued for the first time in this paper, but it is a physical interpretation of an exact bulk formula, not itself a new derivation from the CFT side.

Read this way, (8) says something that would be worth stating even without the word “gravity” anywhere in it: a boundary CFT_2 's semiclassical free energy, expressed as a function of the state's stress tensor at fixed conformal structure, is not an arbitrary function of two variables. It is one fixed function of a single combined variable, and the CMC clock reading is nothing but a bookkeeping device for which value of that variable a given bulk saddle happens to sit at. The dilation in question acts on the stress-tensor phase space itself, at fixed background surface; it is not literally the ordinary dilatation operator of the CFT, which acts on spacetime coordinates and local operators, and we are careful not to conflate the two. The precise object being dilated is better called a Virasoro-coadjoint dilation, in line with the description above.

6.3 One boundary function, two regimes

Corollary 5.5, read through the same dictionary, becomes a statement about $\mathcal{E}$ itself:

$$\begin{aligned} \mathcal{E}(X, T) &= \text{Vol}(X) + \|T\|_{WP}^2 - \frac{1}{2} \mathcal{R}(T) + O(\|T\|^6) \quad (\|T\| \rightarrow 0), \\ \mathcal{E}(X, T) &\longrightarrow \|T\|_1 \quad (\|T\| \rightarrow \infty). \end{aligned}$$

Physically: for a boundary state whose stress tensor is small compared to the natural Weil–Petersson scale, the semiclassical free energy responds quadratically, with the Weil–Petersson metric itself playing the role of the stress tensor's two-point normalisation, corrected at next order by a term set by the curvature of Teichmüller space; this is exactly the behaviour one would expect of the leading terms of an effective action for a weakly self-interacting field on moduli space, and the quadratic coefficient's identification with the Weil–Petersson metric is precisely the content of the Belavin–Knizhnik and Zograf–Takhtajan theorems relating the WP metric to the Liouville action and to the central charge, so it should be regarded as an expected consistency check on the dictionary rather than as new evidence for it. For a heavy state, whose stress tensor is large, the same free energy degenerates into an entirely different, degree-one, Finsler-type functional governed by the extremal-length geometry of the quadratic differential T : the boundary theory, at large enough stress, organises itself around the same singular measured foliations that appear in Thurston's compactification of Teichmüller space, rather than around any Riemannian notion of distance in stress-tensor space at all.

6.4 A sharp, falsifiable prediction

The genuinely new content of the identification lies exactly where the theorem's proof does not reach: the two asymp-

otic regimes above are guaranteed to have the stated leading shape by kinematics on both sides of the duality, quite independently of whether $\mathcal{E}$ and the true CFT stress-tensor effective action coincide beyond leading order. The first order at which the two constructions could actually be distinguished is the quartic curvature term itself, taken beyond the universal coefficient fixed by the central charge, and, more sharply, the sextic term in $\|T\|$, which is not fixed by any symmetry argument known to us on either side. Computing this term for H_1 requires one further iterate of the Kazdan–Warner method used in the proof of Theorem 4.1; computing the corresponding term for the boundary effective action requires an independent semiclassical CFT calculation, for instance along the lines of the large- c Virasoro block or heavy-state stress-tensor techniques developed in the modern literature on $\text{AdS}_3/\text{CFT}_2$. We regard this single, well-posed, and currently unperformed computation as the sharpest test the identification proposed here can currently be given: agreement would be a nontrivial, quantitative confirmation that the exact bulk scaling law and the boundary generating functional are the same object; disagreement would show, just as usefully, exactly where the correspondence between q and the exact quantum stress tensor first breaks down.

6.5 Relation to volume time and $T\bar{T}$

This proposal sits naturally alongside a recent and independent line of work. Blacker, Callebaut, Hergueta, and Ning have shown that York time possesses a canonically conjugate “volume time,” and that the radial canonical formalism built from this pair admits a direct, semiclassical interpretation in terms of a $T\bar{T}$ -deformed CFT, with the CMC-time Hamiltonian generating volume-time translations in exactly the Schrödinger sense familiar from $T\bar{T}$ -deformed holography. That work and the present one are asking closely related questions from opposite ends: theirs establishes, independently of the exact bulk solution used here, that CMC/York time already has a well-defined home in the $T\bar{T}$ literature; ours supplies the exact, closed-form Hamiltonian that this radial evolution is generated by, together with the scaling law that Hamiltonian is forced to obey. Bringing the two together, the exact profile H_1 derived in Section 5 is a natural candidate for the finite-cutoff, $T\bar{T}$ -deformed free energy itself, and the dilation law of Theorem 6.1 would then be read as a statement about how that finite-cutoff free energy depends on the cutoff and on the state jointly. We leave a precise version of this correspondence to future work, but record it here because it suggests the sharp test proposed above is not an isolated numerology exercise but sits inside an active and rapidly developing programme.

6.6 What this is, and is not, claiming

It is worth closing this section with a short list of cautions, in the same spirit as the honesty that made Sections 2 through 5 checkable in the first place. First, the identification of q with the boundary stress tensor is exact only in the sense that holographic renormalisation makes precise; at finite τ it is the statement, established in the literature cited above, that the relevant holographic charges are periods of q , not the claim that every component of q at every value of τ is separately renormalisation-scheme-

independent. Second, everything in this section is a statement about the classical, large- c limit of the CFT; it says nothing, on its own, about $1/c$ corrections, which come from bulk loops that a purely classical construction such as this one cannot see. Third, the dilation acting on T in (8) is a transformation of the stress-tensor phase space at fixed background surface, not the spacetime dilatation generator of the CFT, and we have deliberately avoided writing any equation that would suggest otherwise. Within these boundaries, however, the claim is a genuine and sharply falsifiable one, and we believe it is the right place to end a paper whose entire earlier content is, by contrast, exact.

7 Discussion

Step back from the equations, and the picture that remains is a simple one. A CMC clock in AdS_3 gravity does not hand the theory a new Hamiltonian for every tick; it hands the theory the same Hamiltonian, over and over, read at a radius that grows as the clock runs. The Weil–Petersson geometry near the wormhole’s waist and the Finsler geometry far from it, usually presented as separate features of the subject, are the same fact seen from two distances. And because the momentum this one function depends on is not a bookkeeping variable internal to the bulk, but is exactly, via a dictionary independently established long before this paper, the stress tensor of a two-dimensional conformal field theory living on the boundary, the fact about the wormhole’s breathing becomes, without any further assumption beyond that dictionary, a fact about how the dual theory’s own free energy must be built.

We think this is a satisfying place for the classical part of the story to end, and a promising place for the boundary part to begin. The classical results of Sections 2 through 5, the exact CMC Hamiltonian, its dilation law, the Euler equation it satisfies, and the trigonometric compactification of an unboundedly long clock into a finite angle, do not depend on anything beyond the elliptic theory of a single, well-studied constraint equation, and we would be glad to see them checked, extended, or falsified on their own terms. The holographic reading of Section 6 rests everything additional it claims on one further, independently motivated, and already partially tested identification, and offers, in return, a single sharp number, the sextic coefficient of the boundary effective action, that would settle the matter either way. Very little in quantum gravity offers that combination of ambition and a genuine numerical target; if this one does, whichever way the number comes out, the exercise of asking exactly how a wormhole tells time will have taught us something concrete about the theory living on its edge.

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